

Fig. 3 | Completed model-class background-audit matrix

Background sensitivity is consistent across model architectures.

Target / site	CNN/Mega	LGBM multi	LGBM single	Weis LR
E. coli / Cipro / A-2018	0.82 -> 0.70	0.83 -> 0.64	0.83 -> 0.65	0.73 -> 0.69
E. coli / Cipro / DRIAMS-C	0.75 -> 0.65	0.81 -> 0.64	0.80 -> 0.68	0.63 -> 0.82*
E. coli / Cipro / DRIAMS-D	0.67 -> 0.60	0.72 -> 0.58	0.71 -> 0.59	0.67 -> 0.62
E. coli / Amox-Clav / A-2018	0.65 -> 0.54	0.63 -> 0.48	0.61 -> 0.47	0.56 -> 0.53
E. coli / Amox-Clav / DRIAMS-C	0.54 -> 0.50	0.59 -> 0.52	0.56 -> 0.53	0.63 -> 0.58
E. coli / Amox-Clav / DRIAMS-D	0.56 -> 0.49	0.58 -> 0.45	0.56 -> 0.49	0.65 -> 0.52
S. aureus / Oxa / A-2018	0.84 -> 0.76	0.79 -> 0.69	0.84 -> 0.73	not run
S. aureus / Oxa / DRIAMS-B	0.78 -> 0.86	0.77 -> 0.81	0.82 -> 0.87	0.67 -> 0.67*
S. aureus / Oxa / DRIAMS-C	0.73 -> 0.70	0.66 -> 0.62	0.72 -> 0.73	0.54 -> 0.54

Cells show raw AUC followed by background-centered AUC for CNN/Mega, multi-task LightGBM, single-task LightGBM and Weis/Borgwardt LR rows. CNN and LightGBM use threshold matched support; N/A = no usable S/R oxacillin-labeled S. aureus expanded study panels. Weis/Borgwardt rows are shown separately: exact upstream parity was verified only for the official LR subset, while other Weis-format rows test audit portability rather than serve as pooled benchmarks. Asterisks mark low matched support; retention and valid-stratum counts are reported in the source data.

prediction rows in DRIAMS-D export; 'not run' denotes no corresponding Weis/Borgwardt LR audit row for that target/site.